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Polynomials and Vieta's formulas

Level 3 Senior · Algebra

How the coefficients of a polynomial encode its roots, and how to use remainders, Vieta's formulas and symmetric expressions to answer questions about roots you never find.

By the end you can: find remainders and factors quickly with the remainder and factor theorems; evaluate symmetric expressions in the roots of a polynomial without solving it; use the equation itself to compute power sums and to build polynomials with related roots.

Lesson

Here is a problem that looks unpleasant. The equation $x^3 - 5x^2 + 3x - 7 = 0$ has roots a, b and c. Find $a^2 + b^2 + c^2$.

You could try to solve the cubic. It has no rational roots (the only candidates are ± 1 and ± 7, and none of them works), one real root near 4.68 and two complex roots. Nobody wants to square those. But watch this. The cubic factorises as $(x-a)(x-b)(x-c)$, and expanding that and comparing with the given cubic tells us $a+b+c = 5$ and $ab+bc+ca = 3$. Then

$$a^2 + b^2 + c^2 = (a+b+c)^2 - 2(ab+bc+ca) = 25 - 6 = 19.$$

We never found a single root. That is the spirit of this unit: the coefficients of a polynomial already know a great deal about its roots, and contest problems are set so that you only ever need what the coefficients tell you.

Division, remainders and the factor theorem

Everything rests on polynomial division. If P is a polynomial and D is a non-zero polynomial, there are unique polynomials Q (the quotient) and R (the remainder) with

$$P(x) = D(x)\,Q(x) + R(x), \qquad \deg R < \deg D.$$

This is long division, just as with integers, and "the degree of the remainder is less than the degree of the divisor" plays the role of "the remainder is smaller than the divisor".

> **Remainder theorem.** The remainder when $P(x)$ is divided by $x - k$ is $P(k)$.
> **Factor theorem.** $x - k$ is a factor of $P(x)$ if and only if $P(k) = 0$.

The proof is two lines. Dividing by $x - k$, which has degree 1, leaves a remainder of degree 0, that is a constant r. So $P(x) = (x-k)Q(x) + r$ for every x. Put $x = k$: the first term vanishes and $P(k) = r$. The factor theorem is the case $r = 0$.

Three habits follow. First, dividing by $x + 3$ means $k = -3$, so the remainder is $P(-3)$. Second, when you divide by a quadratic the remainder has the form $px + q$, which has two unknowns, so you need two facts to pin it down. If the quadratic has distinct roots α and β, those facts are $P(\alpha)$ and $P(\beta)$, because the $D(x)Q(x)$ term vanishes at both. Third, if you know the values of a polynomial at several points, subtract something simple that takes the same values. The difference has known roots, so the factor theorem pins it down once you know its degree and leading coefficient.

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Vieta's formulas

Suppose $P(x) = ax^3 + bx^2 + cx + d$ with $a \neq 0$ has roots r, s, t, where repeated roots are listed as often as they occur and complex roots are allowed. By the factor theorem, used three times, $P(x) = a(x - r)(x - s)(x - t)$. Expand the right-hand side:

$$a(x-r)(x-s)(x-t) = a\big(x^3 - (r+s+t)x^2 + (rs+st+tr)x - rst\big).$$

Now compare coefficients with $ax^3 + bx^2 + cx + d$.

Vieta's formulas. If $ax^3 + bx^2 + cx + d$ has roots r, s, t, then

$$r + s + t = -\frac{b}{a}, \qquad rs + st + tr = \frac{c}{a}, \qquad rst = -\frac{d}{a}.$$

For a quadratic $ax^2 + bx + c$ with roots r, s we get $r + s = -\frac{b}{a}$ and $rs = \frac{c}{a}$. In degree n, the sum of the products of the roots taken k at a time equals $(-1)^k$ times the coefficient of x^{n-k}, divided by the leading coefficient.

The signs alternate, starting with a minus, and you always divide by the leading coefficient. I will write $e_1 = r + s + t$, $e_2 = rs + st + tr$ and $e_3 = rst$. These are the *elementary symmetric polynomials* of the roots. (That a polynomial of degree n has exactly n complex roots, counted with multiplicity, is the Fundamental Theorem of Algebra. You may quote it.)

Symmetric expressions

An expression in r, s, t is *symmetric* if swapping any two of the letters leaves it unchanged, like $r^2 + s^2 + t^2$ or $\frac{1}{r} + \frac{1}{s} + \frac{1}{t}$. A classical theorem says that every symmetric polynomial in the roots can be written as a polynomial in e_1, e_2, e_3, so every such expression can be computed from the coefficients alone. In practice you need a small toolkit, each line of which you can check by expanding:

- $r^2 + s^2 + t^2 = e_1^2 - 2e_2$;
- $\frac{1}{r} + \frac{1}{s} + \frac{1}{t} = \frac{e_2}{e_3}$;
- $r^3 + s^3 + t^3 = e_1^3 - 3e_1e_2 + 3e_3$;
- $(k - r)(k - s)(k - t) = \frac{P(k)}{a}$ for any number k.

The last line is the one people forget, and it is often the fastest of all. It is nothing more than the factorisation $P(x) = a(x - r)(x - s)(x - t)$ evaluated at $x = k$.

Higher power sums $p_n = r^n + s^n + t^n$ come from the equation itself. For a monic cubic $x^3 + bx^2 + cx + d$, each root satisfies $r^3 + br^2 + cr + d = 0$. Multiply by r^{n-3} and add the three versions of this equation:

$$p_n + b\,p_{n-1} + c\,p_{n-2} + d\,p_{n-3} = 0 \qquad (n \geq 3),$$

where $p_0 = 3$. This is the friendliest form of Newton's identities: it turns power sums into a recurrence.

New roots from old

Sometimes you want a polynomial whose roots are related to the old ones. If P has roots r_i, then $P(x - 2)$ has roots $r_i + 2$, $P(x/3)$ has roots $3r_i$, and reversing the coefficients of P gives a polynomial with roots $\frac{1}{r_i}$ (provided no root is 0). Once you have the new polynomial, Vieta applies to it as usual.

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Example 1. The roots of $x^2 - 7x + 3 = 0$ are p and q. Find $\frac{p}{q} + \frac{q}{p}$.

Solution.
The first instinct is the quadratic formula, which gives $\frac{7 \pm \sqrt{37}}{2}$. You can push that through, but dividing surds by surds is slow and invites errors. Notice instead that the target does not change if we swap p and q, so it must be expressible through $p + q = 7$ and $pq = 3$.
Combining the fractions, $\frac{p}{q} + \frac{q}{p} = \frac{p^2+q^2}{pq} = \frac{(p+q)^2-2pq}{pq} = \frac{49-6}{3} = \frac{43}{3}$.
The lesson: when the target is symmetric, rewrite it in terms of the sum and product before doing anything else.

Example 2. When $P(x)$ is divided by $x^2 - 4$ the remainder is $3x + 1$, and when $P(x)$ is divided by $x - 3$ the remainder is 15. Find the remainder when $P(x)$ is divided by $(x^2 - 4)(x - 3)$.

Solution.
The divisor is a cubic, so the remainder $R(x)$ has degree at most 2. The direct approach is to write $R(x) = ux^2 + vx + w$ and find three equations. We know $P(2) = 7$, $P(-2) = -5$ and $P(3) = 15$, and R agrees with P at the roots $2, -2, 3$ of the divisor, so this system can be solved. It works, but it is clumsy, and there is a better way.
Write $P(x) = (x^2 - 4)(x - 3)Q(x) + R(x)$. The first term is a multiple of $x^2 - 4$, so P and R leave the same remainder on division by $x^2 - 4$, namely $3x + 1$. Since R has degree at most 2, its quotient on division by $x^2 - 4$ is a constant, so $R(x) = c(x^2 - 4) + 3x + 1$ for some number c. Now there is only one unknown.
Putting $x = 3$ gives $15 = P(3) = R(3) = 5c + 10$, so $c = 1$ and the remainder is $x^2 + 3x - 3$. As a check, at $x = 2$ this gives $4 + 6 - 3 = 7$, which matches $3 \cdot 2 + 1$.

Example 3. The roots of $x^3 - 4x^2 + x + 5 = 0$ are a, b and c. Find $(a + b)(b + c)(c + a)$.

Solution.
You could expand. The product is $a^2b + ab^2 + b^2c + bc^2 + c^2a + ca^2 + 2abc$, and with some effort that equals $e_1e_2 - e_3$. Here $e_1 = 4$, $e_2 = 1$ and $e_3 = -5$, so the answer is $4 + 5 = 9$. Fine, but it needs an identity you may not remember under pressure.
The better observation is that each bracket is the sum of all the roots with one left out. Since $a + b + c = 4$, we have $a + b = 4 - c$, and similarly for the others. So the product is $(4 - a)(4 - b)(4 - c)$, which by the last line of the toolkit is $P(4) = 64 - 64 + 4 + 5 = 9$.
The lesson: a bracket containing all but one root should be rewritten as e_1 minus that root, and then a product of such brackets is a value of P.

Example 4. Let a, b, c be the roots of $x^3 - x - 1 = 0$. Find $a^5 + b^5 + c^5$.

Solution.
Expanding $(a + b + c)^5$ and subtracting the unwanted terms is hopeless. The roots themselves are unpleasant: one is about 1.3247 and the other two are complex. So let the equation do the work. Each root satisfies $r^3 = r + 1$, and multiplying by r^{n-3} gives $r^n = r^{n-2} + r^{n-3}$. Adding this over the three roots gives $p_n = p_{n-2} + p_{n-3}$.
We need starting values. Clearly $p_0 = 3$, and $p_1 = e_1 = 0$ because there is no x^2 term. Also $e_2 = -1$, so $p_2 = e_1^2 - 2e_2 = 2$. Then $p_3 = p_1 + p_0 = 3$, $p_4 = p_2 + p_1 = 2$ and $p_5 = p_3 + p_2 = 5$.
As a sanity check, $1.3247^5 \approx 4.08$, so the complex pair should contribute about 0.92, which is a perfectly believable real number for a conjugate pair. The answer is 5. The lesson: when the exponent is large, use the equation to reduce powers.

Common traps.

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- Sign errors. For ax^3+bx^2+cx+d the sum of the roots is $-\frac{b}{a}$, and the product is $-\frac{d}{a}$, whereas for a quadratic the product is $+\frac{c}{a}$. If in doubt, expand $(x-r)(x-s)(x-t)$ again.
- Forgetting the leading coefficient. For $2x^2-6x+1$ the product of the roots is $\frac{1}{2}$, not 1.
- Missing terms. In x^3-7x+6 the coefficient of x^2 is 0, so the roots add up to 0.
- Vieta counts every root, complex and repeated ones included. If the question asks for the sum of the *real* roots or of the *distinct* roots, find out which roots those are first.
- A negative value for $r^2+s^2+t^2$ is not an error. It just means some roots are not real.
- The remainder on division by $x+k$ is $P(-k)$, not $P(k)$.

How to spot this in a contest

The signals are reliable. A question names the roots of a polynomial but the roots are ugly, or the polynomial has no obvious factor. The thing you are asked for is symmetric in the roots. You are told remainders, or the values $P(1), P(2), \ldots$ of a polynomial of known degree, in which case look at $P(x)$ minus something simple. A product such as $(2-r)(2-s)(2-t)$ appears, which is a value of P in disguise. In every case the plan is the same: do not solve, translate into e_1, e_2, e_3 or into values of P, and compute.

On the SMC these questions are quick wins once spotted, so read the coefficients before trying anything clever. On the AMC 12 and the AIME the same tools appear in disguise: a substitution to tidy the coefficients, a root that can be guessed, or two polynomials whose roots are related. If a calculation is getting long, ask whether you have missed a symmetric shortcut.

Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆ The roots of $2x^2-6x+1=0$ are r and s. Find r^2+s^2.

2. ⋆⋆ Find the remainder when $x^{2026}+3x^{101}-5$ is divided by x^2-1.

3. ⋆⋆ The equation $x^3-6x^2+kx+10=0$ has three real roots that form an arithmetic progression. Find k.

4. ⋆⋆⋆ A polynomial $P(x)$ of degree 4 with leading coefficient 1 satisfies $P(1)=2$, $P(2)=5$, $P(3)=10$ and $P(4)=17$. Find $P(5)+P(0)$.

5. ⋆⋆⋆ Let a, b and c be the roots of $x^3-2x^2+3x-4=0$. Find $(a^2+1)(b^2+1)(c^2+1)$.

6. ⋆⋆⋆⋆ Let a, b and c be the roots of $x^3-4x+1=0$. Find

$$\frac{a^2}{a+2}+\frac{b^2}{b+2}+\frac{c^2}{c+2}.$$

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

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1. UKMT SMC 2015 Question 2 ⋆
2. AMC 10A 2003 Problem 18 ⋆
3. AHSME 1999 Problem 17 ⋆⋆
4. UKMT SMC 2024 Question 11 ⋆⋆
5. UKMT SMC 2023 Question 18 ⋆⋆
6. AMC 12A 2022 Problem 15 ⋆⋆
7. AMC 12B 2004 Problem 17 ⋆⋆⋆
8. UKMT SMC 2022 Question 19 ⋆⋆⋆
9. AIME 1996 Problem 5 ⋆⋆⋆
10. AIME II 2008 Problem 7 ⋆⋆⋆
11. AIME I 2014 Problem 9 ⋆⋆⋆⋆
12. AIME II 2014 Problem 5 ⋆⋆⋆⋆

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Hints

1. Write $r^2 + s^2$ using $r + s$ and rs, and remember to divide by the leading coefficient.
2. The remainder has the form $px + q$. Which two values of x make $x^2 - 1$ vanish?
3. Call the roots $m - d$, m and $m + d$. What does the sum of the roots tell you about m?
4. The given values look like $n^2 + 1$. What can you say about $P(x) - (x^2 + 1)$?
5. Factorise $a^2 + 1$ using i, then remember that $(k - a)(k - b)(k - c) = P(k)$ for a monic cubic P.
6. First divide: $\frac{a^2}{a+2} = a - 2 + \frac{4}{a+2}$. Then find a cubic whose roots are $a + 2$, $b + 2$ and $c + 2$.

Solutions

Problem 1

Answer: 8

By Vieta's formulas, $r + s = \frac{6}{2} = 3$ and $rs = \frac{1}{2}$. Then $r^2 + s^2 = (r + s)^2 - 2rs = 9 - 1 = 8$.

Problem 2

Answer: $3x - 4$

Write $x^{2026} + 3x^{101} - 5 = (x^2 - 1)Q(x) + px + q$. Putting $x = 1$ gives $1 + 3 - 5 = p + q$, so $p + q = -1$. Putting $x = -1$, and using that 2026 is even and 101 is odd, gives $1 - 3 - 5 = -p + q$, so $q - p = -7$. Adding the two equations gives $2q = -8$, so $q = -4$ and $p = 3$. The remainder is $3x - 4$.

Another way to see it: when working modulo $x^2 - 1$ we may replace x^2 by 1, so $x^{2026} = (x^2)^{1013}$ becomes 1 and $x^{101} = x(x^2)^{50}$ becomes x, leaving $1 + 3x - 5$.

Problem 3

Answer: 3

Let the roots be $m - d$, m and $m + d$. Their sum is $3m$, and by Vieta it equals 6, so $m = 2$ and 2 is a root. Substituting $x = 2$ gives $8 - 24 + 2k + 10 = 0$, so $k = 3$.

We should check that this value really works. Indeed $x^3 - 6x^2 + 3x + 10 = (x - 2)(x^2 - 4x - 5) = (x - 2)(x + 1)(x - 5)$, whose roots -1, 2, 5 are real and form an arithmetic progression. So $k = 3$.

Problem 4

Answer: 75

The given values are $1^2 + 1$, $2^2 + 1$, $3^2 + 1$ and $4^2 + 1$. So the polynomial $Q(x) = P(x) - (x^2 + 1)$ vanishes at $x = 1, 2, 3, 4$. Subtracting a quadratic does not change the x^4 term, so Q has degree 4 and leading coefficient 1. By the factor theorem, $Q(x) = (x - 1)(x - 2)(x - 3)(x - 4)$.

Hence $P(5) = 26 + 4 \cdot 3 \cdot 2 \cdot 1 = 50$ and $P(0) = 1 + (-1)(-2)(-3)(-4) = 25$, so $P(5) + P(0) = 75$. Note that the tempting guess $P(x) = x^2 + 1$ is not allowed, because P must have degree 4.

Problem 5

Answer: 8

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Write $P(x) = x^3 - 2x^2 + 3x - 4 = (x-a)(x-b)(x-c)$. Since $a^2+1 = (i-a)(-i-a)$, the product we want is $(i-a)(i-b)(i-c)\cdot(-i-a)(-i-b)(-i-c) = P(i)\,P(-i)$. Now $P(i) = -i+2+3i-4 = -2+2i$ and $P(-i) = i+2-3i-4 = -2-2i$, so the answer is $(-2+2i)(-2-2i) = 4+4 = 8$.

If you prefer to avoid complex numbers, expand instead: the product equals $a^2b^2c^2 + (a^2b^2 + b^2c^2 + c^2a^2) + (a^2+b^2+c^2) + 1$, which is $e_3^2 + (e_2^2 - 2e_1e_3) + (e_1^2 - 2e_2) + 1$. With $e_1 = 2$, $e_2 = 3$ and $e_3 = 4$ this is $16 - 7 - 2 + 1 = 8$.

Problem 6

Answer: -38

No root equals -2, since $P(-2) = -8+8+1 = 1 \neq 0$, so every fraction makes sense. Polynomial division gives $a^2 = (a-2)(a+2) + 4$, so $\frac{a^2}{a+2} = a - 2 + \frac{4}{a+2}$. Adding the three versions, the required sum is $(a+b+c) - 6 + 4S$, where $S = \frac{1}{a+2} + \frac{1}{b+2} + \frac{1}{c+2}$. By Vieta $a+b+c = 0$, because there is no x^2 term.

To find S, note that $y = x+2$ turns the roots a, b, c into $a+2, b+2, c+2$. These are the roots of $P(y-2) = (y-2)^3 - 4(y-2) + 1 = y^3 - 6y^2 + 8y + 1$. For this cubic, the sum of the reciprocals of the roots is $\frac{e_2}{e_3} = \frac{8}{-1} = -8$. So $S = -8$ and the required sum is $0 - 6 + 4(-8) = -38$.

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